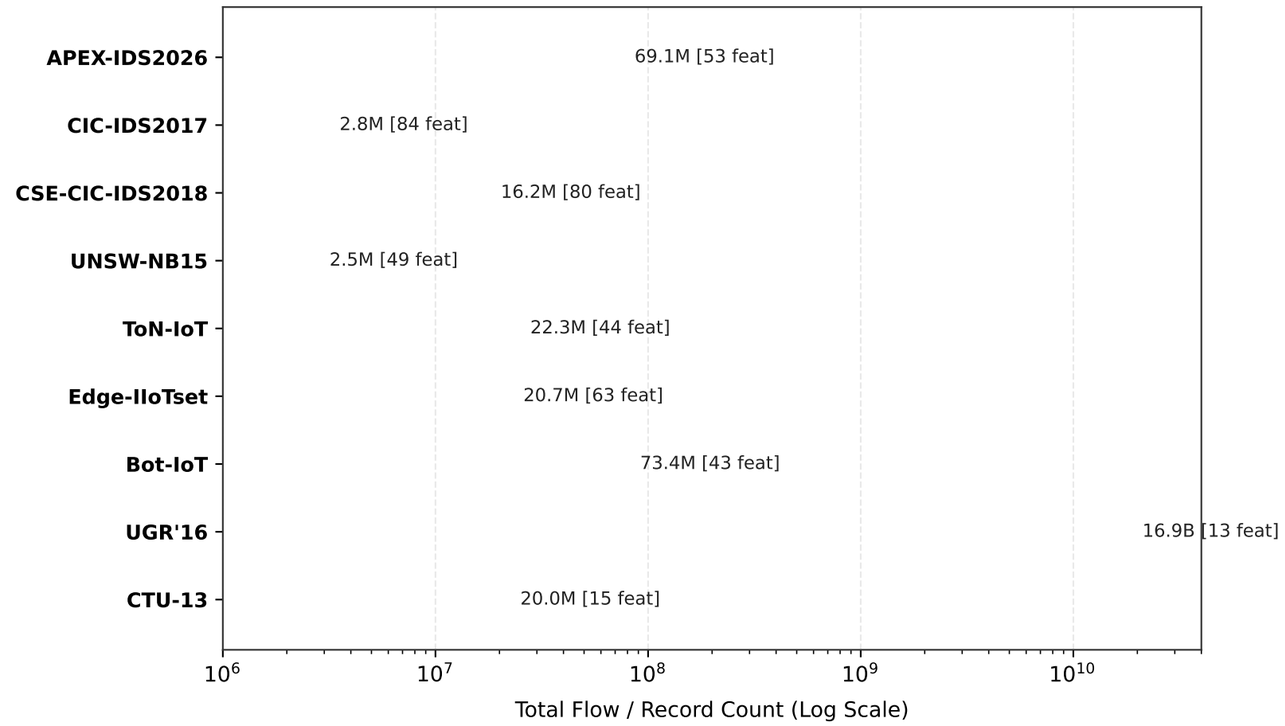
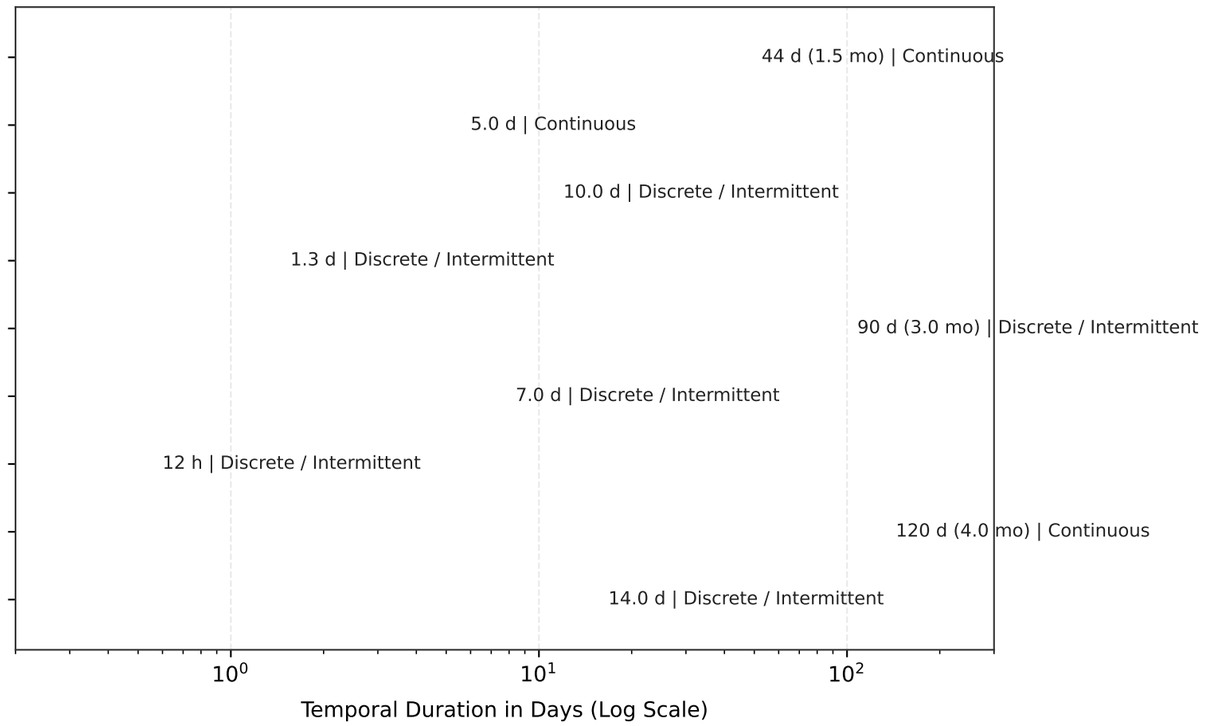
(A) Dataset Scale and Feature Dimensions



(B) Temporal Scope and Capture Continuity



(C) Temporal & Network-Topology Properties

	Timestamps Documented	Endpoint IDs (Src/Dst IPs)	Chronological Ordering	Repeated Observations	Attack Labels	Real Traffic Evidence
APEX-IDS2026	● Yes	● Yes	● Yes	● Yes	● Yes	● Yes
CIC-IDS2017	● Yes	● Yes	● Yes	● Yes	● Yes	◑ Partial
CSE-CIC-IDS2018	● Yes	◑ Partial	● Yes	◑ Partial	● Yes	◑ Partial
UNSW-NB15	◑ Partial	◑ Partial	○ No	○ No	● Yes	◑ Partial
ToN-IoT	● Yes	● Yes	◑ Partial	◑ Partial	● Yes	◑ Partial
Edge-IIoTset	● Yes	● Yes	◑ Partial	◑ Partial	● Yes	◑ Partial
Bot-IoT	● Yes	● Yes	◑ Partial	○ No	● Yes	○ No
UGR'16	● Yes	◑ Partial	● Yes	● Yes	◑ Partial	● Yes
CTU-13	● Yes	● Yes	◑ Partial	◑ Partial	● Yes	● Yes

(D) CTG Analytical Suitability Matrix

	Temporal Graph Construction	Endpoint-Level Resolution	Chronological Train/Val/Test	Attack-Label Integrity	Dataset Scale & Realism
APEX-IDS2026	● High	● High	● High	● High	● High
CIC-IDS2017	● High	● High	● High	● High	◑ Mod.
CSE-CIC-IDS2018	◑ Mod.	◑ Mod.	◑ Mod.	● High	● High
UNSW-NB15	○ Low	○ Low	○ Low	● High	◑ Mod.
ToN-IoT	◑ Mod.	● High	◑ Mod.	● High	● High
Edge-IIoTset	◑ Mod.	● High	◑ Mod.	● High	● High
Bot-IoT	○ Low	● High	○ Low	● High	● High
UGR'16	● High	◌ Unver.	● High	◑ Mod.	● High
CTU-13	◑ Mod.	● High	◑ Mod.	● High	● High

Primary Benchmark (APEX-IDS2026) | Secondary Benchmark (CIC-IDS2017) | Status: Yes / High | Status: Partial / Moderate | Status: No / Low